# Evaluation of Bioinformatics Tools for Carbapenem-Resistant Organism Genomic Surveillance and Cluster Detection

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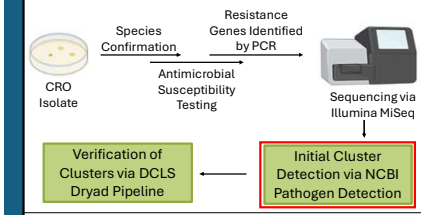
**Abstract** Carbapenem-resistant organisms (CROs) are bacteria associated with healthcare-acquired infections that have developed a resistance to carbapenems and are a growing public health threat. The Virginia Division of Consolidated Laboratory Services (DCLS) aims to develop a new, internal method that quickly and accurately identifies genomic clusters for CROs. Towards this goal, the ability of clustering methods PopPUNK and floc to accurately assign CRO isolates to previously identified outbreak clusters was evaluated. The sensitivity and specificity for each method was calculated, and it was found that floc maximized sensitivity while maintaining reasonable specificity in comparison to PopPUNK. From these results, floc is the optimal clustering method for the carbapenem-resistant organisms tested.

## Background

Carbapenems are a class of antibiotics reserved to treat infections caused by multidrug-resistant microbes. Carbapenem-resistant organisms (CROs) are microbes that have developed a resistance to carbapenems and are a growing public health threat. The Virginia Division of Consolidated Laboratory Services (DCLS) is the public health laboratory for the state of Virginia, responsible for performing genomic surveillance for CROs and reporting any identified genomic clusters to the Virginia Department of Health (VDH). Current methods for genomic surveillance and cluster detection of CROs are time intensive and reliant on external resources.

### Current Cluster Identification Pipeline




### Proposed Change

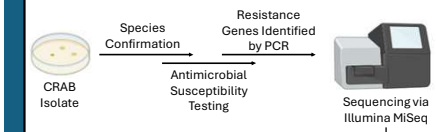
Replace **NCBI Pathogen Detection** with **PopPUNK or floc**

DCLS aims to develop a new, internal method that quickly and accurately identifies genomic clusters. Towards this goal, bioinformatics tools PopPUNK and floc were tested for their ability to replace NCBI Pathogen Detection as the initial clustering step in the pipeline. Utilizing carbapenem-resistant *Acinetobacter baumannii* (CRAB) as an initial test case, each tool was evaluated on its ability to correctly assign each of 674 CRAB isolates to previously-identified outbreak clusters.

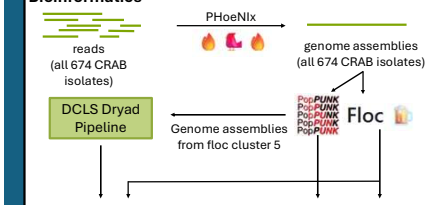
With a new genomic surveillance and cluster detection method that implements the tools tested, DCLS will be able to report CRO surveillance data to VDH more quickly to aid epidemiologic investigations. This improvement will ultimately lead to faster outbreak detection and better-informed public health response, thereby reducing the spread of CROs.

## Methods

### Sample Acquisition, Processing, Sequencing




### Bioinformatics




**Neighbor-Joining Tree Generation followed by Visualization in Microreact (Figures 2 and 3)**

**Sensitivity and Specificity (Table 1)**
**Sensitivity:** ability of a method to correctly identify an isolate as a member of a DCLS cluster
**Specificity:** ability of a method to correctly reject isolates outside of a DCLS cluster

## Results

### Table 1. floc and PopPUNK

**Main Finding: floc maximizes sensitivity while maintaining specificity**

| Method | Average Sensitivity | Average Specificity |
|---|---|---|
| floc | 1 | 0.8901 |
| PopPUNK | 1 | 0.5608 |
| PopPUNK+PopPIPE | 0.7388 | 0.9770 |

**Table 1**. Table depicting average sensitivity and specificity for clustering methods floc and PopPUNK. Two PopPUNK methods were tested, with and without PopPIPE subclustering to see effect on scores.

### Figure 2. DCLS Dryad Pipeline vs. floc

**Main Finding: floc successfully clusters genetically related isolates**

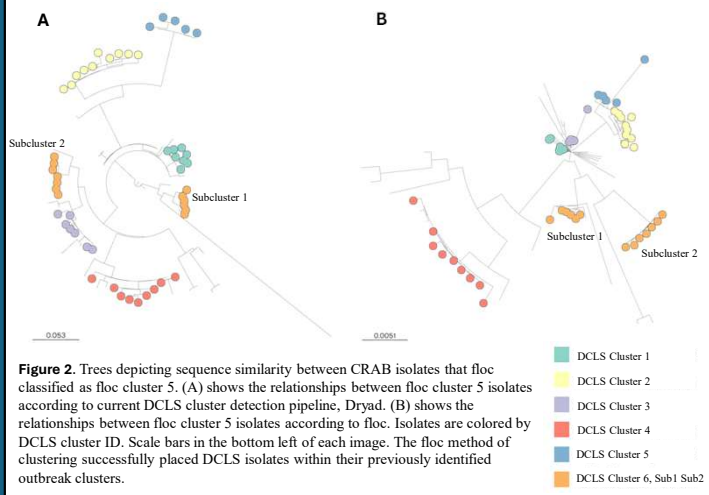



**Figure 2**. Trees depicting sequence similarity between CRAB isolates that floc classified as floc cluster 5. (A) shows the relationships between floc cluster 5 isolates according to current DCLS cluster detection pipeline, Dryad. (B) shows the relationships between floc cluster 5 isolates according to floc. Isolates are colored by DCLS cluster ID. Scale bars in the bottom left of each image. The floc method of clustering successfully placed DCLS isolates within their previously identified outbreak clusters.

### Figure 3. PopPUNK+PopPIPE vs floc

**Main Finding: PopPUNK+PopPIPE incorrectly separates genetically related isolates**

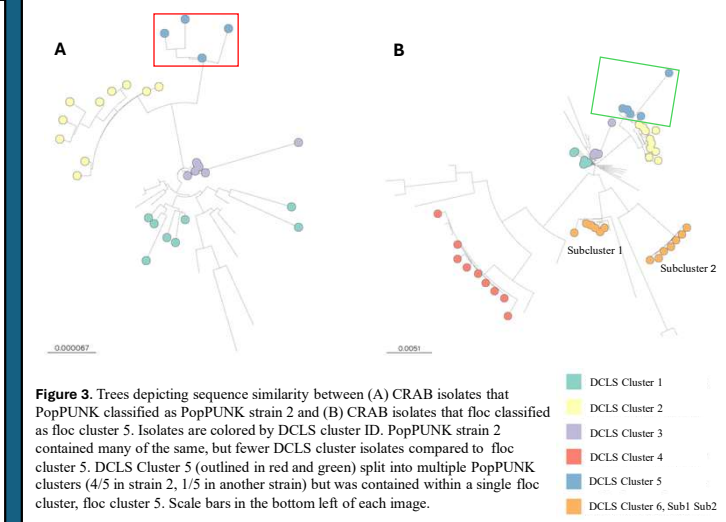


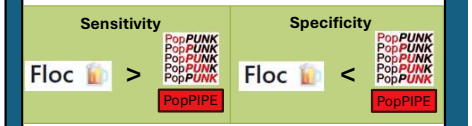

**Figure 3**. Trees depicting sequence similarity between (A) CRAB isolates that PopPUNK classified as PopPUNK strain 2 and (B) CRAB isolates that floc classified as floc cluster 5. Isolates are colored by DCLS cluster ID. PopPUNK strain 2 contained many of the same, but fewer DCLS cluster isolates compared to floc cluster 5. DCLS Cluster 5 (outlined in red and green) split into multiple PopPUNK clusters (4/5 in strain 2, 1/5 in another strain) but was contained within a single floc cluster, floc cluster 5. Scale bars in the bottom left of each image.

## Discussion

### CRAB Clustering Output

Comparing the outputs of PopPUNK and floc, there was a clear tradeoff between sensitivity and specificity. When comparing floc and PopPUNK without PopPIPE subclustering, there was comparable sensitivity between the two methods, but PopPUNK had decreased specificity (Table 1). Therefore, if floc and PopPUNK were followed by a reference-based SNP calling pipeline, the larger PopPUNK clusters would require increased time and computational resources to analyze in comparison to the smaller floc clusters.

PopPUNK specificity was increased by including the PopPIPE subclustering workflow, but sensitivity suffered as a result (Table 1). This meant that PopPUNK+PopPIPE was more likely than floc to incorrectly separate related isolates into multiple PopPUNK clusters. This was observed in Figure 3, where PopPUNK+PopPIPE separated a DCLS Cluster 5 isolate that was only 5-12 SNPs away from all other isolates in DCLS Cluster 5.




**Overall, floc was the method that maximized sensitivity while maintaining reasonable specificity when compared to PopPUNK with and without PopPIPE subclustering.**

### Preliminary Carbapenem-Resistant *Klebsiella pneumoniae* Clustering Output

Both PopPUNK methods and floc were tested on carbapenem-resistant *Klebsiella pneumoniae*. Whereas CRAB in Virginia is highly clonal, *Klebsiella pneumoniae* is not. Initially, it was suspected that clustering performance may differ and floc may not be the optimal choice for both species. However, it was observed that floc and PopPUNK+PopPIPE had equal sensitivity and floc had increased specificity (Supplementary Table 1). **Therefore, it was concluded that floc is also the optimal clustering method for carbapenem-resistant *Klebsiella pneumoniae*.**

### Next Steps

Floc + Reference-based SNP Calling Pipeline

1. Combine floc with a reference-based SNP calling pipeline to get SNP clusters with smaller SNP ranges and see how the output compares to previous DCLS reports
2. Implement and validate new methods to DCLS standards

### Impact

Faster Reporting to VDH → Faster Outbreak Detection → Reduced Spread of CROs

**Supplementary Materials & References**

**Contact Us!**

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